

Muhammad Abdullah

Lahore, Pakistan



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PROFILE

Software Engineering student at UCP building full-stack web, mobile, and distributed backend systems. Co-founded MedAlerto, a healthcare platform that manages over 100+ patient records using React Native, Node.js, MongoDB, and Spring Boot.

EXPERIENCE

- **Co-Founder & Full Stack developer** 11/2025 – Present

MedAlerto | [Live Site](#) ↗

 - Shipped a production Node.js/Express REST API and React Native mobile app for an AI-powered patient management platform handling 100+ active patient records used daily by clinicians.
 - Designed a JWT role-based multi-entity REST API with MongoDB and PostgreSQL, supporting patient records, scheduling, billing, AI analysis, and PDF prescription/report generation, with compound indexing for optimized dashboard queries.
 - Integrated Python AI/ML microservices, WhatsApp Business API, SendGrid, and Docker-based cloud deployment, automating medication and appointment reminders by 80% and enabling scalable, high-availability operation across web and mobile.
- **Junior Full-Stack Developer (Contract Based)** 09/2025 – 02/2026

Dev Verx | Lahore, Pakistan

 - Delivered full-stack features with Express.js REST APIs and React.js, independently developing and integrating backend and frontend layers.
 - Optimized MongoDB aggregation pipelines and query performance by eliminating N+1 patterns and implementing compound indexes, reducing API latency by 30%.
 - Architected scalable document and relational database schemas with referential integrity and index coverage, while implementing centralized error handling and input validation to prevent production crashes.

PROJECTS

- Well's Merry (Freelance Project)** View ↗

 - Developed a responsive e-commerce site for a natural hair-oil brand using React, Next.js, and Tailwind CSS, featuring product storytelling and conversion-focused layouts.
 - Created and implemented an interactive hair quiz that routes visitors to personalized product recommendations, improving discovery-to-purchase conversion on desktop and mobile.

React Next.js Tailwind CSS Express.js Node.js MongoDB
- Real-Time Messaging Service** View ↗

 - Engineered a bi-directional WebSocket system with Socket.IO, tracking user connection states and delivering room-targeted broadcasts using O(1) in-memory lookups.
 - Implemented Redis Pub/Sub as a message broker to horizontally scale WebSocket connections across multiple server nodes.
 - Stored chat history in MongoDB with cursor-based pagination, avoiding full-collection scans and maintaining efficient retrieval as data volume increased.

Node.js Socket.IO MongoDB Redis Pub/Sub React.js

Distributed Banking Service

[View](#)

- * Constructed a stateless Spring Boot REST API for account creation, deposits, withdrawals, and transfers, implementing idempotency checks to prevent duplicate transactions.
- * Added Redis caching for balance lookups, reducing database round-trips by 40% on high-read endpoints.
- * Modeled a PostgreSQL schema with JPA/Hibernate and compound indexes on (user_id, created_at) for efficient paginated transaction history retrieval.
- * Established JWT authentication, refresh-token rotation, and role-based access control (RBAC) to securely separate user and admin privileges.

Spring Boot Java PostgreSQL Redis JWT

Chess Engine

[View](#)

- * Programmed full move-validation for all piece types with check/checkmate detection; encoded game state as a 64-cell array with O(1) positional lookup per piece.
- * Applied Command pattern for move history enabling undo/redo via O(1) stack operations; separated game-state, rules, and rendering into distinct architectural layers.

Java OOP Multithreading Data Structures

EDUCATION

University of Central Punjab (UCP)

2023 – 2027

Bachelor of Science in Software Engineering (BSSE)

CGPA: 3.73

SKILLS

Languages	JavaScript	TypeScript	Java	C++					
Frontend	React.js	Next.js	TailwindCSS	Bootstrap	HTML5	CSS3			
Mobile Node.js	ReactNative	Cross-platformiOS&Android							
Stack	Node.js	Express.js	WebSockets(Socket.IO)	JWT	MiddlewarePipelines	RESTful APIDesign			
	MicroservicesArchitecture	JWTAuthentication	OAuth2.0						
Java / Spring	SpringBoot	SpringMVC	SpringSecurity	SpringDataJPA	Hibernate	Maven	RESTAPIs		
Databases	MongoDB	PostgreSQL	MySQL	Redis(caching,pub/sub)	JPA/Hibernate	QueryOptimisation			
Messaging	ApacheKafka	Event-drivenarchitecture	Producer/Consumerpatterns						
Integrations DevOps	WhatsAppBusinessAPI	SendGrid	PDFGeneration	PythonMLmicroservices					
& Tools	Git	GitHub	Postman	Docker	Vercel	CloudFlare	Render	Netlify	AWS

ACHIEVEMENTS

Healthcare Platform:

2026

- * Deployed and actively maintain a live healthcare platform used daily by doctors and medical professionals.

Full-Stack Development:

2023-26

- * Built and shipped 10+ applications across React Native, React/Next.js, Node.js, and Spring Boot, including distributed backend systems.

DSA:

2023-2026

- * Solved 100+ LeetCode problems across arrays, hashmaps, trees, and dynamic programming.